

Design for SiRNA

21-23 nt dsRNA

GC% slightly < 50%

Perfect complimentary to target mRNA

Targeting 3' UTR works better than 5' UTR

NGS assembles three categories based on graph (repeat)

NGS assemblers have been organized into three categories, based on graphs: • **Overlap-Layout-Consensus (OLC)** methods rely on an overlap graph.

Graph Algorithms for Assembly • the **de Bruijn Graph (DBG) methods** use some form of K-mer graph • The **greedy graph algorithms** may use OLC or DBG

Hamiltonian Path • The Hamiltonian path in a graph is a path that includes each vertex of the graph once and only once • At the end, of course, the circuit must return to the starting vertex.

Eulerian Path • An Eulerian path is a path that visits every edge of the graph once and only once. • It can end on a vertex different from the one on which it began.

Eulerian Circuit • An Eulerian path which begins and ends on the same vertex. • It starts and ends on the same vertex.

Three tools to find gene prediction in eukaryotes

Genie

Fgenes

Genscan

Examples of some important hormones

Testosterone • Estrogen • Growth Hormone (GH) • Follicle Stimulating Hormone (FSH) • Thyroid Hormone (TH) • Melanocyte Stimulating Hormone (MSH) • Anti Diuretic Hormone (ADH) • Prolactin • are generally small proteins, many hormones. e.g. Oxytocin Occurs females and stimulates uterine.

We sequence genome to understand the mechanism of which type of disease

With the help of three dimensional analyses of proteins we have found that HIV protease is the enzyme which is responsible for AIDS.

Biological importance of proteins (repeat)

Cellular Function • Structural importance • Enzymes • Hormones • Transport protein • Messenger protein • Antibodies • Plasma protein • Protein in diet

Factors for quality assembled genome

Following factors may be important in planning an Assembly project;

Size of Genome • Repeat content • Related species Assembled • Strategies for performing Assembly

The sequencing approaches and assembly strategies are interdependent

Significance of genome

Genie is approximately 60- 75% accurate on eukaryotic genomes.

Challenges face for human genome

Size • Polymorphism • Repeats - smaller repeats are technically difficult to sequence • Some DNA sequence are repeated all over the genome. Relies on cloning (Some regions are difficult to clone - Heterochromatin). • Some sequences rearrange or are deleted when cloned.

Write down the classification of transposable elements

IS elements , transposons, eukaryotic TES,

Advances of third generation sequencing

Less sample preparation (no PCR) • No amplification • no PCR errors • fewer contamination issues • no GC-bias • analyze every sample (unPCRable / unclonable) • analyze low quality DNA (museum, archeological, forensics samples) • Absolute quantification • Sequence RNA directly

Write about alpha helix structure of protein

The simplest arrangement which a polypeptide chain could assume with its rigid peptide bonds is a helical structure, which Pauling and Corey called **the α -helix**.

OLC advantage

Suitable for large size genomes • Works for both short and long reads • OLC approach is capable of handling NGS data • Generate correct order of contigs • OLC recognizes repeats but may collapse them into one – ABBBCBBD => ABCBD

Maxam gilbert DNA sequencing method

Maxam-Gilbert's Method • Walter Gilbert and his student Allan Maxam at Harvard also developed sequencing methods, including one for **"DNA sequencing by chemical degradation"**. • Chemical

modification followed by cleavage • 5' end of the DNA to be sequenced is labelled (P32) • Helps in screening

Spotted glass arrays

• Uses cDNA, Oligonucleotide, protein, antibody • Robotically spotted cDNAs or Oligonucleotides • Printed on Nylon, Plastic, or Glass microscope slide

Greedy graph algorithm based assemblers

• SSAKE • VCAKE • SHARCGS

Work of greedy graph

Greedy Graph algorithms works as follows: • Compare all reads or contigs in a pairwise fashion to identify overlapping sequences • Merge the sequences that overlap each other the best. Repeat step 2 until no more sequences can be merged, or the remaining overlaps conflict with existing contigs.

Advantages of single molecule sequence

2009 : 1st Helicos single molecule sequencer : Helicos Genetic Analyser System

2011 : 1st Ion Torrent Next Generation

Sequencer : PGM 1st Pacific Biosciences single molecule

sequencer : PacBio RS Systems

2012 : Oxford Nanopore Technologies demonstrates ultra long single molecule reads

Clone by clone and whole genome shotgun sequencing difference repeat

Clone by clone

Also known as hierarchical shotgun sequencing or map based shotgun sequencing. The target DNA is first analysed by clone based physical mapping methods • Individual clones that together span the region of interest are selected and subjected to shotgun sequencing.

Whole genome shotgun sequencing

Whole genome is broken down, cloned into vectors and then sequenced. The DNA is broken up randomly into numerous small segments, which are sequenced.

Convergent pathway and divergent pathway

Divergence of a path into two branches that afterwards join together again into one path • Possible causes include sequencing errors toward the middle of a read, and by polymorphisms in the target.

Inverse definition than for the bubbles, two paths converge into one, that later diverges again into two separate paths • Possible causes are repeats in the target genome.

SiRNA

RISC is a large (~500- kDa) RNA multiprotein complex, which triggers mRNA degradation in response to siRNA. Unwinding of doublestranded siRNA by ATP independent helicase.

4 ways of finding quality control

Filter the graph of erroneous occurrences (i.e., bubbles, spurs, cycles) convergences or divergences. Nodes that are unambiguously connected by an edge are merged together.

Dicer and dorsha

Dicer : • Dicer or helicase with RNase motif, is an enzyme of RNase III family • In humans, it is encoded by the DICER1 gene. • Dicer cleaves double-stranded RNA (dsRNA) and pre-microRNA into short doublestranded RNA fragments called small interfering RNA and microRNA. • It is able to digest dsRNA into uniformly sized small RNAs (siRNA) • Dicer family proteins are ATP-dependent nucleases. • RNase III enzyme acts as a dimer. • Dicer homologs exist in many organisms including C. elegans, Drosophila, yeast and humans.

• Drosha is a class 2 ribonuclease III enzyme that in humans is encoded by the gene DROSHA (formerly RNASEN) gene. • The RNase III drosha is the core nuclease that executes the initiation step of microRNA (miRNA) processing in the nucleus.

• Dicer is an enzyme that is part of the RNase III family • Drosha belongs to class 2 ribonuclease III enzyme family.

Biogenesis of miRNA

Small non-coding double stranded RNAs • Approximately 19-22 nucleotides long • miRNA originates with ssRNA that forms a hairpin secondary structure. Originate from capped & polyadenylated full length precursors (pri-miRNA).